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Reduced molecular representations and their role in protein structure prediction[1]

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Abstract

The technique of nonlinear mapping is shown to permit two-dimensional representations of protein structure. The resulting diagrams retain distance geometry information to a distance matrix root mean square (dmrms) accuracy of about 3.0 Å, and in a striking manner maintain the fold and secondary structural features of the three-dimensional entity. By assuming a consistent carbon alpha to carbon alpha distance, we make a further reduction to a one-dimensional angle plot which itself retains distance relationships to an accuracy not appreciably worse than the two-dimensional plot.

It is possible to go from one dimension to two, and hence to three without complications arising from chirality by using a sign convention and inspecting the overall three-dimensional structure which may need inverting.

There are numerous possible applications of this approach but the hope is that it will provide a basis for going from a one-dimensional gene sequence to a tertiary protein structure. © 1997 Elsevier Science B.V.

Keywords: Protein structure; Nonlinear mapping

1. Introduction

Gene sequences are abundantly available. Protein structure has been believed to be coded in the sequence ever since Anfinsen's work on renaturation and denaturation [1]. Despite this, going from the one-dimensional sequence of nucleic acids or amino acids to the three-dimensional protein structures remains a tantalizingly unsolved problem. The majority of attempts are based on a logic which may be summarized as: primary structure → secondary structure → tertiary structure.

Here we start with an alternative basis, that of: one dimension ↔ two dimensions ↔ three dimensions. We will seek new types of representation that may simplify the problem. The key step is a novel two-dimensional representation [2].

2. A two-dimensional representation of proteins

If we start with a protein of known three-dimensional structure then the geometric information may be summarized in a distance matrix. This is a matrix of all C_α–C_α distances in the protein and has utility in identifying homologies and comparing shapes [3]. The two-dimensional representation we seek is that two-dimensional representation which most closely retains distance information. The mapping from 3D to 2D is performed by using an algorithm originally developed by Sammon [4] for

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multi-dimensional data analysis. It involves minimization of the difference between the distance matrix of the original structure and that of a novel two-dimensional structure.

A protein of N amino acids can be described by N three-dimensional vectors, $P_i(x_i, y_i, z_i)$, $i = 1,\cdots,N$. P_i represents the alpha carbon coordinates of residue i. We define N corresponding vectors in two-dimensional space: $W_i = (w_{i1}, w_{i2})$, $i = 1,\cdots, N$. Initially the two-dimensional coordinates are randomly assigned. Let d^*_{ij} be the distance between residue i and residue j in the three-dimensional structure. That is

$$d^*_{ij} = [(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2]^{1/2} \tag{1}$$

Let d_{ij} be the corresponding inter-amino acid distance in the two-dimensional representation.

We can now define an error, $E(m)$ which describes how well the interamino acid distances in the two-dimensional representation compare to those in the three-dimensional structure.

$$E(m) = \frac{1}{\sum\limits_{i<j}^{N}[d^*_{ij}]} \sum_{i<j}^{N} \frac{[d^*_{ij} - d_{ij}(m)]^2}{d^*_{ij}} \tag{2}$$

This error can be minimized by an iterative process. In Eq. (2), m labels the iteration number. The two-dimensional vectors, W_i are modified to minimize the error by a steepest descent method. The new components of W_p are given by

$$w_{pq}(m+1) = w_{pq}(m) - \eta\Delta_{pq}(m) \tag{3}$$

where η is a learning rate parameter set at 0.3 and

$$\Delta_{pq}(m) = \frac{\partial E(m)}{\partial w_{pq}(m)} \Bigg/ \left|\frac{\partial^2 E(m)}{\partial w_{pq}(m)^2}\right| \tag{4}$$

Eq. (3) is further constrained so that none of the two-dimensional vectors, W_i become identical. This would cause problems in the determination of the partial derivatives. The partial derivatives are readily determined as follows

$$\frac{\partial E}{\partial w_{pq}} = \frac{-2}{\sum\limits_{i<j}^{N} d^*_{ij}} \sum_{\substack{j=1 \\ j \neq p}}^{N} \left[\frac{d^*_{pj} - d_{pj}}{d_{pj} d^*_{pj}}\right](w_{pq} - w_{jq}) \tag{5}$$

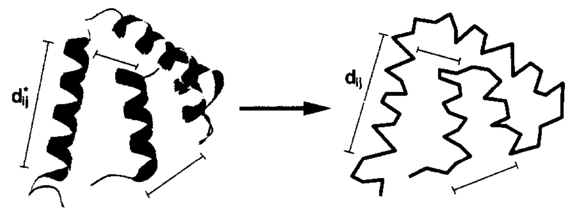

Fig. 1. Nonlinear mapping of the three-dimensional structure of rabbit uteroglobin. Mapping preserves as much as possible the inter-C_α distances. For the two-dimensional representation on the right, the chain order is shown by joining each of the minimized two-dimensional coordinates with a line.

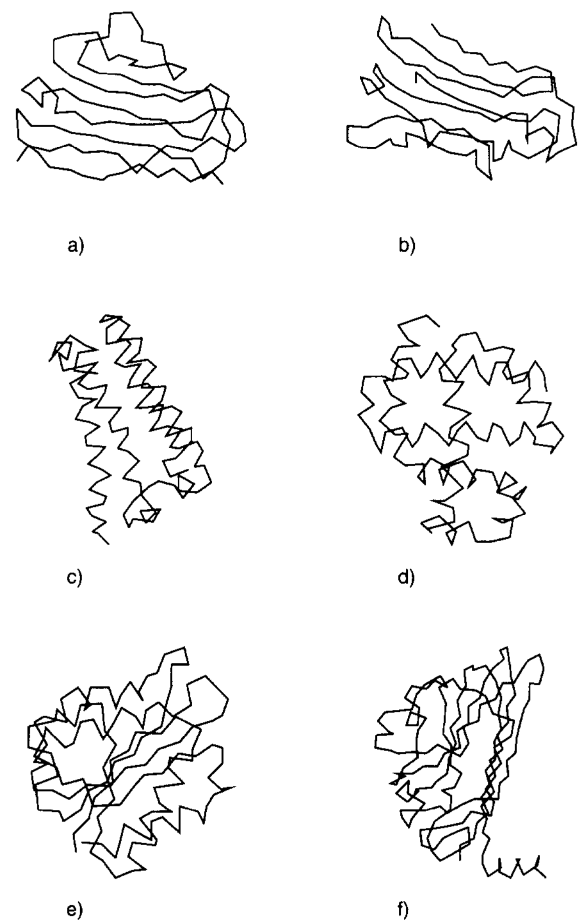

Fig. 2. Nonlinear maps of example proteins. (a) Bence–Jones Immunoglobulin REI variable portion (1REI); (b) Prealbumin (human plasma) (2PAB); (c) Cytochrome c (prime) (2CCY); (d) Hemoglobin (erythocruorin, deoxy) (1ECD); (e) Flavodoxin (1FX1); (f) Satellite tobacco necrosis virus (2STV).

Table 1
Distance matrix root mean square deviation (dmrms) calculated for the nonlinear maps of proteins specified by the Brookhaven code. dmrms = $1/N\sqrt{\sum_{i\neq j}^{n}[d_{ij}^{*}-d_{ij}]^2}$, where d_{ij} is the distance between residue i and residue j in the approximate structure, d_{ij}^{*} is the corresponding distance in the X-ray structure, and N is the total number of residues in the chain. (Distances between residues are measured as distances between C_α atoms.) The root mean square deviation for distances greater than 12 Å (dmrms12) has also been calculated

Code	Unit	Res.	Helix	Sheet	Coil	dmrms (Å)	dmrms12 (Å)
1REI	A	107	0	60	47	3.1	2.3
1PFC		111	4	35	72	2.3	1.7
2PAB	A	114	7	61	46	3.3	2.8
2RHE		114	6	58	50	3.4	2.6
2MCP	H	222	6	117	99	2.5	1.8
2STV		184	10	91	83	2.9	2.6
2MLT	A	26	23	0	3	0.6	0.5
1PPT		36	19	0	17	0.7	0.7
2CCY	A	127	86	2	39	2.7	2.0
2HMQ	A	113	77	0	36	2.8	2.0
1ECD		136	99	0	37	2.9	2.2
1MBD		153	112	0	41	2.8	2.1
1FX1		147	55	33	59	3.2	2.3
8API	A	340	105	120	115	4.6	3.8

$$\frac{\partial^2 E}{\partial w_{pq}^2}=\frac{-2}{\sum_{i<j}^{N} d_{ij}^{*}}\sum_{\substack{j=1\\ j\neq p}}^{N}\frac{1}{d_{pj}d_{pj}^{*}}\times\left[(d_{pj}^{*}-d_{pj})-\frac{(w_{pq}-w_{jq})^2}{d_{pj}}\left(1+\frac{d_{pj}^{*}-d_{pj}}{d_{pj}}\right)\right] \quad (6)$$

Repeated evaluation of Eq. (2) followed by modification of the two-dimensional coordinates by Eq. (3) produces a nonlinear two-dimensional mapping of the three-dimensional protein structure. The resulting two-dimensional coordinates have no direct physical significance. However, these coordinates can be plotted to obtain a two-dimensional representation of protein structure where the distances between coordinates reflect the distances between C_α's in the original structure. See Fig. 1.

In Fig. 2, a selection of 2D plots are presented. Original structures were taken from the Brookhaven data bank. It is quite striking how secondary structural features are clearly discernable, as are the essentials of folding.

Just how good these representations are in quantitative terms is highlighted in Table 1.

These distances are good enough to use as input to distance geometry programs which can recreate a 3D structure from the 2D diagrams. In general, distance geometry has problems with chiral structures. However, in this case there are so many distances that only two solutions are possible in going from 2D to 3D; the true structure or its mirror image. They may be distinguished by eye from a three-dimensional display (helices must be right-handed), so if necessary the structure can be inverted which is trivial.

Having demonstrated that these novel 2D structures retain geometrical relationships and can be used as a step to 3D models we now turn our attention to a 1D geometric model which again must retain distance information to a high precision and permit stepping in both directions between 1D and 2D without chirality problems.

3. A one-dimensional representation of proteins

As is clear from Fig. 1 and Fig. 2, our two-dimensional representations look like a series of jagged lines with each vertex being a C_α atom and the points connected by C_α–C_α bonds. It cannot be overemphasized that these figures are not projections, but the best possible 2D plots constrained to preserve as much information as possible from native distance matrices. If we now further make the reasonable assumption that the C_α–C_α distances are constant

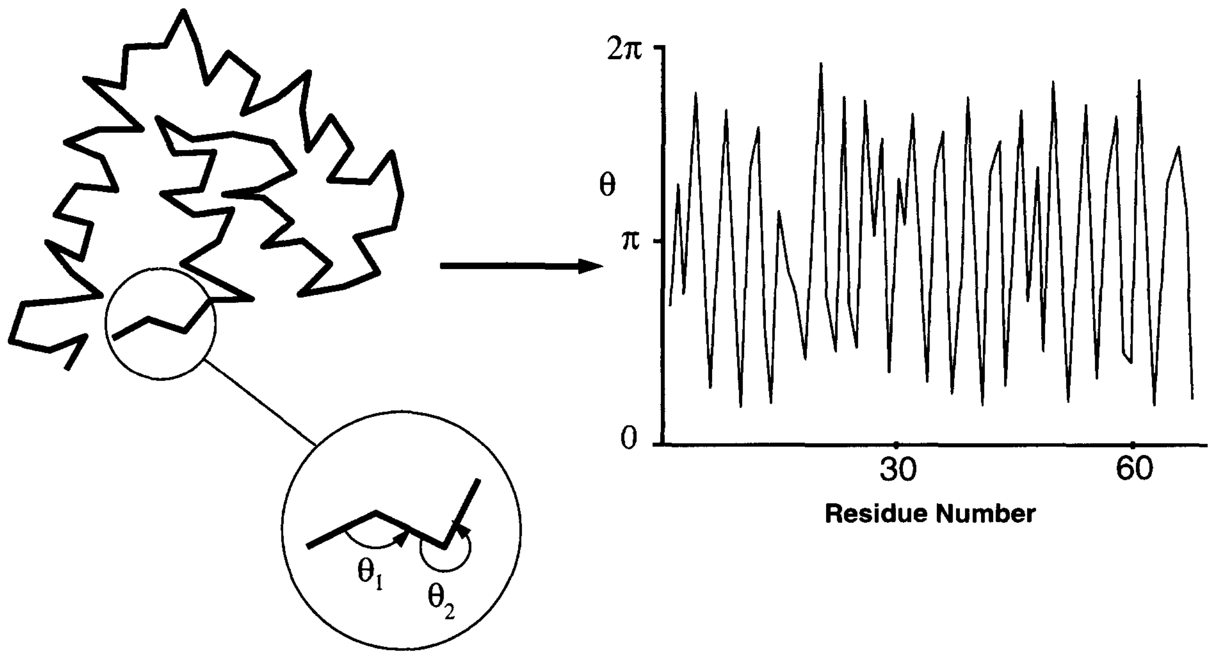

Fig. 3. A 2D representation of protein structure, with neighbouring C_α to C_α distances fixed to a constant value, can be characterized by a 1D spectrum of angles, θ.

and equal to 3.801 Å, then the angles between adjacent bonds, θ, provide the basis of a one-dimensional spectrum of angles from which the 2D and hence 3D structures may be derived [5]. The angles must be defined using a sign convention which avoids ambiguity about which way one turns when progressing along the chain. We have chosen to proceed from the N-terminus towards the C-terminus, measuring the angle at each vertex in an anticlockwise direction, as illustrated in Fig. 3.

Table 2
The distance matrix root mean square deviation (dmrms) has been calculated in Å for both the two-dimensional map (2D) and the one-dimensional spectrum (1D) obtained by fixing all C_α–C_α distances to 3.801A Å

Code	Unit	Res.	Helix	Sheet	Coil	2D	1D
1REI	A	107	0	60	47	3.1	3.3
1PFC		111	4	35	72	2.3	2.6
2PAB	A	114	7	61	46	3.3	3.6
2RHE		114	6	58	50	3.4	3.6
2MCP	H	222	6	117	99	2.5	2.7
2STV		184	10	91	83	2.9	3.1
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1ECD		136	99	0	37	2.9	3.1
1MBD		153	112	0	41	2.8	3.0
1FX1		147	55	33	59	3.2	3.4
8AP1	A	340	105	120	115	4.6	4.8
Average		1930	609	576	731	3.1	3.3

Table 2 indicates that the dmrms deviation of these 1D representations from the three-dimensional structures is just over 3 Å for a wide range of proteins from the Brookhaven data bank.

4. Conclusions

We have shown that there are two- and one-dimensional representations of protein structure that retain distance information to an acceptable degree. It is also possible to move in both directions between 1D, 2D and 3D representations so that the 1D diagram can summarize tertiary structures. We envisage a number of possible applications for both the 2D and 1D diagrams, but potentially the most exciting would be to try to go from the 1D sequence to the 1D distance diagram using neural networks. Attempts to go from sequence to secondary structure using neural networks have hitherto proved disappointing, but predicting 1D structure from 1D sequence seems a more promising route. Networks are highly susceptible to the patterns that are presented to them. The one-dimensional representation of protein structure outlined here, or even simplifications thereof could be just what is needed.

Acknowledgements

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